



System Architecture



Client Layer

Web browsers, mobile apps, API clients

HTTP/HTTPS requests



Django REST API

Contract analysis endpoints, rate limiting, validation

Python 3.11 + DRF



AI Analysis Engine

OpenAI GPT-4.1 integration, industry-specific prompts

Contract intelligence



Document Processor

OCR, text extraction, format detection

Tesseract + PIL



PostgreSQL Database

Analysis results, user tracking, metadata

ACID compliance



Redis Cache

Rate limiting, session management

In-memory storage



Data Processing Pipeline

1

File Upload & Validation

Check file size (≤10MB), format (PDF/DOCX/TXT/Images), industry type

2

Rate Limit Check

Verify IP hasn't exceeded 5 daily uploads, increment counter

3

Document Type Detection

Determine if PDF is searchable or scanned, route to appropriate processor

4

Text Extraction

PyPDF2 for searchable PDFs, OCR pipeline for scanned documents/images

5

AI Analysis

Send extracted text to GPT-4.1 with industry-specific prompts

6

Result Processing

Parse JSON response, extract risk score, store in database

7

Response Generation

Format analysis results, calculate processing time, return to client



API Request Flow

Client Request



Rate Limiting



File Validation



Document Processing



AI Analysis



Database Storage



JSON Response



Technology Stack

Backend Framework

Django 5.2.4
Django REST Framework 3.16.0

AI Engine

OpenAI GPT-4.1
Custom Prompting

Document Processing

Tesseract OCR
PIL + pdf2image

Database

PostgreSQL
Redis Cache

Deployment

Docker
Gunicorn

Documentation

Swagger UI
ReDoc



API Endpoints

POST

/api/v1/contracts/

Upload and analyze contract document

GET

/api/v1/contracts/{id}/

Retrieve analysis result by ID

GET

/api/v1/contracts/

List all contract analyses (paginated)

GET

/api/v1/health/

API health check endpoint

GET

/api/v1/rate-limit/

Check rate limit status for current IP



Rate Limiting System



Daily Limits

5 documents per IP address per day

Resets at midnight UTC • IP-based tracking • Redis storage



OCR Processing Pipeline

PDF Upload



Searchable Check



pdf2image Conversion



PIL Preprocessing



Tesseract OCR



Text Output



Image Preprocessing Steps:

- Resize large images (max 2000px)
- Convert to grayscale
- Enhance contrast (1.5x)
- Apply sharpening filter
- Convert to binary (black/white)
- Memory optimization with garbage collection



Security & Performance Features



File Security

Size limits (10MB upload, 100MB processing)

Extension validation

Timeout protection (30s)



Performance

Memory optimization

Processing time tracking

Image resizing for large files



Production Security

SSL enforcement

CSRF protection

Environment variables



Monitoring

Health check endpoints

Error logging

Status tracking